

MAT21-3

Multivariable Analysis

Gradient, Hessian, critical points, and gradient descent

Extending calculus to several variables to analyse local behaviour and justify gradient-based optimization.

Albert School — 22 hours

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1 Functions of several variables

A **function of several variables** assigns a real number to a point of $\mathbb{R}^2$ or $\mathbb{R}^3$. We write $f : D \rightarrow \mathbb{R}$ with $D \subseteq \mathbb{R}^n$ ($n = 2$ or 3) its **domain**: the set of points at which the defining expression makes sense. Domains are treated operationally, with no formal topology.

Worked example. $f(x, y) = \sqrt{1 - x^2 - y^2}$ has domain $D = \{(x, y) : x^2 + y^2 \leq 1\}$, the closed unit disk. $g(x, y) = \ln(xy)$ has domain $D = \{(x, y) : xy > 0\}$.

1.1 Graphical interpretation

The **graph** of $f : D \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}$ is the surface

$$\{(x, y, f(x, y)) : (x, y) \in D\} \subseteq \mathbb{R}^3.$$

A **level set** (level curve in $\mathbb{R}^2$, level surface in $\mathbb{R}^3$) is the set of points where f takes a fixed value c :

$$\{(x, y) : f(x, y) = c\}.$$

Standard examples include $f(x, y) = x^2 + y^2$ (level curves are circles) and $f(x, y) = x^2 - y^2$ (level curves are hyperbolas).

2 Partial derivatives

Definition 1 Let $f : D \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}$ and $(a, b) \in D$. The **partial derivatives** of f at (a, b) are

$$\frac{\partial f}{\partial x}(a, b) = \lim_{h \rightarrow 0} \frac{f(a+h, b) - f(a, b)}{h}, \quad \frac{\partial f}{\partial y}(a, b) = \lim_{h \rightarrow 0} \frac{f(a, b+h) - f(a, b)}{h},$$

when these limits exist. The definition extends to $\mathbb{R}^3$ and beyond, one variable at a time.

In practice $\partial f / \partial x$ is computed by differentiating with respect to x while treating the other variables as constants.

Worked example. For $f(x, y) = x^2y + \sin(y)$,

$$\frac{\partial f}{\partial x} = 2xy, \quad \frac{\partial f}{\partial y} = x^2 + \cos(y).$$

2.1 Geometric interpretation

$\frac{\partial f}{\partial x}(a, b)$ is the slope of the cross-section of the graph cut by the plane $y = b$, measured in the x -direction; $\frac{\partial f}{\partial y}(a, b)$ is the slope of the cross-section cut by $x = a$.

3 Differentiability

Definition 2 f is **differentiable** at (a, b) if there is a linear map L with

$$f(a+h, b+k) = f(a, b) + L(h, k) + o(\|(h, k)\|)$$

as $(h, k) \rightarrow (0, 0)$. When it exists, $L(h, k) = \frac{\partial f}{\partial x}(a, b)h + \frac{\partial f}{\partial y}(a, b)k$.

The practical criterion is the **C^1 condition**.

Theorem 3 ($C^1 \Rightarrow$ differentiable $\Rightarrow$ continuous) If the partial derivatives of f exist and are continuous on a neighbourhood of (a, b) (i.e. f is C^1 there), then f is differentiable at (a, b) ; and if f is differentiable at (a, b) , then f is continuous at (a, b) .

Neither converse holds.

Common pitfall The existence of both partial derivatives does **not** imply differentiability, nor even continuity. For

$$f(x, y) = \begin{cases} \frac{xy}{x^2+y^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases},$$

both partials exist at the origin (they equal 0), yet f is not continuous there, hence not differentiable.

4 Gradient

Definition 4 The **gradient** of a differentiable f at a point is the vector assembled from its partial derivatives:

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right) \quad (\text{and similarly in } \mathbb{R}^3 \setminus).$$

4.1 Directional derivatives

Definition 5 Let u be a unit vector. The **directional derivative** of f at a point p in the direction u is

$$D_u f(p) = \lim_{h \rightarrow 0} \frac{f(p + hu) - f(p)}{h}.$$

Proposition 6 If f is differentiable at p , then for every unit vector u ,

$$D_u f(p) = \nabla f(p) \cdot u.$$

4.2 Steepest ascent

Since $D_u f(p) = \nabla f(p) \cdot u = \|\nabla f(p)\| \cos \theta$, the directional derivative is largest when u points along $\nabla f(p)$. The gradient therefore points in the **direction of steepest ascent**, and its magnitude $\|\nabla f(p)\|$ is the maximal rate of increase.

5 Chain rule and gradient orthogonality

Theorem 7 (Chain rule along a curve) Let f be differentiable and let $\gamma(t) = (x(t), y(t))$ be a differentiable curve. Then $t \mapsto f(\gamma(t))$ is differentiable and

$$\frac{d}{dt} f(\gamma(t)) = \nabla f(\gamma(t)) \cdot \gamma'(t).$$

5.1 Orthogonality to level curves

Proposition 8 The gradient $\nabla f(p)$ is orthogonal to the level curve of f through p .

Proof. Let γ parametrize the level curve through $p = \gamma(t_0)$, so $f(\gamma(t)) = c$ is constant. Differentiating with the chain rule, $0 = \nabla f(\gamma(t)) \cdot \gamma'(t)$. At t_0 , the tangent vector $\gamma'(t_0)$ to the level curve satisfies $\nabla f(p) \cdot \gamma'(t_0) = 0$. $\square$

5.2 Tangent line and tangent plane

For a level curve $f(x, y) = c$ through (a, b) , the **tangent line** is the line through (a, b) orthogonal to $\nabla f(a, b)$:

$$\frac{\partial f}{\partial x}(a, b)(x - a) + \frac{\partial f}{\partial y}(a, b)(y - b) = 0.$$

For the graph $z = f(x, y)$ of a differentiable function, the **tangent plane** at (a, b) is

$$z = f(a, b) + \frac{\partial f}{\partial x}(a, b)(x - a) + \frac{\partial f}{\partial y}(a, b)(y - b).$$

6 Hessian matrix

Definition 9 The **second-order partial derivatives** of f are the partial derivatives of its partial derivatives, written

$$\frac{\partial^2 f}{\partial x^2}, \quad \frac{\partial^2 f}{\partial y^2}, \quad \frac{\partial^2 f}{\partial x \partial y}, \quad \frac{\partial^2 f}{\partial y \partial x}.$$

The **Hessian matrix** of f at a point is

$$H_f = \begin{pmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} \end{pmatrix}.$$

Theorem 10 (Schwarz) If the second-order partial derivatives of f are continuous, then the mixed partials are equal:

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}.$$

In particular the Hessian is symmetric.

6.1 Second-order Taylor expansion

For a C^2 function f , the **second-order Taylor expansion** at a point p , in terms of the displacement $h = (h_1, h_2)$, is

$$f(p+h) = f(p) + \nabla f(p) \cdot h + \frac{1}{2} h^T H_f(p) h + o(\|h\|^2).$$

At a critical point ($\nabla f(p) = 0$) the linear term vanishes and the quadratic term $\frac{1}{2} h^T H_f(p) h$ governs the local behaviour.

7 Critical point classification

Definition 11 A **critical point** of f is a point where $\nabla f = 0$, i.e. a simultaneous solution of

$$\frac{\partial f}{\partial x} = 0, \quad \frac{\partial f}{\partial y} = 0.$$

7.1 Monge criterion

Write, at a critical point,

$$r = \frac{\partial^2 f}{\partial x^2}, \quad s = \frac{\partial^2 f}{\partial x \partial y}, \quad t = \frac{\partial^2 f}{\partial y^2}, \quad D = rt - s^2 = \det H_f.$$

Theorem 12 (Monge criterion) At a critical point with $D = rt - s^2$:

- if $D > 0$ and $r > 0$, it is a **local minimum**;
- if $D > 0$ and $r < 0$, it is a **local maximum**;
- if $D < 0$, it is a **saddle point**;

- if $D = 0$, the test is inconclusive.

7.2 Eigenvalue interpretation

The Hessian is symmetric, so it has real eigenvalues. At a critical point:

- both eigenvalues positive (H_f positive definite) $\rightarrow$ local minimum;
- both eigenvalues negative (H_f negative definite) $\rightarrow$ local maximum;
- eigenvalues of opposite sign $\rightarrow$ saddle point.

Since $D = \det H_f$ is the product of the eigenvalues and $r + t = \text{tr } H_f$ their sum, the Monge criterion is the two-variable reading of this eigenvalue test.

8 Gradient descent

Definition 13 Gradient descent minimizes f by the iteration

$$x_{n+1} = x_n - \alpha \nabla f(x_n),$$

starting from an initial point x_0 , where $\alpha > 0$ is the **step size** (learning rate).

Each step moves in the direction of steepest **descent** $-\nabla f(x_n)$. The step size controls the move length: too small makes progress slow, too large can overshoot and prevent convergence. Where the iteration converges, the limit satisfies $\nabla f = 0$, i.e. it is a **critical point** of f .

9 Conditioning and convergence rate

Near a critical point the local behaviour of f is set by the Hessian eigenvalues. For gradient descent on a quadratic with Hessian eigenvalues $\lambda_1 \geq \dots \geq \lambda_n > 0$, convergence is governed by the **condition number**

$$\kappa = \frac{\lambda_1}{\lambda_n}.$$

When κ is large (an **ill-conditioned** Hessian), the level sets are highly elongated: a step size small enough to avoid overshooting the steep directions makes little progress along the shallow ones, so convergence is slow. When $\kappa \approx 1$ the eigenvalues are comparable and convergence is fast.

10 Local versus global extrema

A **local** minimum (resp. maximum) is least (resp. greatest) only on a neighbourhood; a **global** extremum is so over the whole domain. The criteria above (Monge, eigenvalues) detect local extrema only.

Definition 14 f is **convex** on a convex domain if for all points p, q and $t \in [0, 1]$,

$$f(tp + (1 - t)q) \leq tf(p) + (1 - t)f(q).$$

For a C^2 function, convexity is equivalent to the Hessian being positive semidefinite everywhere.

Proposition 15 If f is convex, then every local minimum is a **global** minimum. In particular a critical point of a convex function is a global minimum.

11 Application to a machine-learning cost function

The tools above analyse a simple ML **cost function** f of its parameters:

1. compute the gradient ∇f and solve $\nabla f = 0$ for the critical point(s);
2. classify each critical point with the Monge criterion or the Hessian eigenvalues;
3. run gradient descent $x_{n+1} = x_n - \alpha \nabla f(x_n)$ and describe its convergence, using the Hessian eigenvalues to anticipate the rate.

When the cost function is convex, the critical point found is the global minimum, so gradient descent converges to the optimal parameters.

12 Vector-valued functions and the Jacobian (general knowledge)

Definition 16 Let $F : \mathbb{R}^n \rightarrow \mathbb{R}^m$, $F = (F_1, \dots, F_m)$. The **Jacobian matrix** of F at a point is the $m \times n$ matrix of all first-order partials

$$J_F = \left(\frac{\partial F_i}{\partial x_j} \right)_{1 \leq i \leq m, 1 \leq j \leq n}.$$

For a scalar function ($m = 1$) the Jacobian is the row vector ∇f .

Theorem 17 (Multivariable chain rule) For composable differentiable maps F and G , the Jacobian of the composition is the product of the Jacobians:

$$J_{G \circ F}(p) = J_G(F(p)) J_F(p).$$

This statement is the prerequisite tool assumed by downstream Deep Learning Mathematics (backpropagation) and Optimization.